



BORIS BORDEAUX

Curriculum Vitae

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I aim to enable non-experts to design complex structures.

Education

- 2021–2025 **PhD in Computer Science**, *Automatic Design of Lacunar Fractal Structures*, Université Bourgogne Europe (UBE), Laboratoire d'Informatique de Bourgogne (LIB), Dijon, France.
PhD Supervisor: Christian GENTIL, Full Professor, Université Bourgogne Europe
Co-supervisor: Lionel GARNIER, Associate Professor (HDR), Université Bourgogne Europe
Thesis Committee:
MORIN Géraldine Full Professor, University of Toulouse Chair
BAC Alexandra Associate Professor (HDR), Aix-Marseille University Reviewer
PELTIER Samuel Full Professor, University of Poitiers Reviewer
ROHMER Damien Full Professor, Institut Polytechnique de Paris Examiner
RAFFIN Romain Full Professor, Université Bourgogne Europe Examiner
RASSKIN Iván Associate Professor, Aix-Marseille University Guest Member
- 2020–2021 **Master's Degree in Image and Artificial Intelligence**, UBE, Dijon, France.
Double-degree program with the engineering school
- 2018–2021 **Engineering Degree in Computer Science**, UBE, Polytech Dijon, France.
Software and Knowledge Engineering Track
- 2016–2018 **University Diploma of Technology (DUT) in Computer Science**, UBE, Dijon, France.

Professional Experience

Positions in Higher Education and Research

- 2026–present **Postdoctoral Researcher**, Cornell Tech, Matter of Tech Lab, New York, NY, USA.
Research on stress-based lightweight structures and multi-stream fractal heat exchangers
- 2024–2025 **Teaching and Research Fellow (ATER)**, UBE, LIB, Dijon, France.
Teaching at the department of Computer Science, Electronics and Mechanics
- 2021–2024 **PhD Student in Computer Science**, UBE, LIB, Dijon, France.
Teaching duties at the department of Computer Science, Electronics and Mechanics

Internships

- 2021 **Web Development (6 months)**, Alteca, Dijon, France.
Web application providing real-time attendance of points of interest (cameras + AI)
- 2020 **Research (3 months)**, I3Mainz, Mainz, Germany.
Unsupervised segmentation for semantic analysis of point clouds
- 2018 **Database (3 months)**, Howden Solyvent-Ventec, Chalon-sur-Saône, France.
Database creation and task automation in VBA

Publications

Journal Articles

- 2026 **Computer-Aided Design of Lacunar Fractal Polygons**, *Computer Aided Geometric Design*, volume 126, page 102529, <https://doi.org/10.1016/j.cagd.2026.102529>.
Boris Bordeaux, Christian Gentil, Iván Rasskin, Lionel Garnier.
- 2024 **Automatic construction of fractal structures with locally controlled lacunarity**, *Journal of WSCG*, volume 32, pages 1–12. Union Agency, Science Press, <https://doi.org/10.24132/JWSCG.2024.1>. Best conference papers from WSCG are published in this journal.
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Thesis

- 2025 **Conception automatique de structures fractales**, *PhD Thesis*, UBE, <https://doi.org/10.70675/2439b786z7e3ez4e09z9751za698e35ee55f>.
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Talks

International Peer-Reviewed Conference

- June 2024 **Automatic construction of fractal structures with locally controlled lacunarity**, WSCG 2024, Pilsen, Czech Republic, <http://wscg.zcu.cz/wscg2024/program.html>. The paper presented at this conference appeared in the *Journal of WSCG*.
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National Conferences without Peer Review

- October 2024 **Design of lacunar fractal structures**, *Geometry & Computing 2024*, Luminy, France, Workgroup meeting presentation, <https://geocomp2024.sciencesconf.org/>.
Boris Bordeaux, Iván Rasskin, Christian Gentil, Lionel Garnier
- October 2023 **Representation of polytopal sphere packings using the BC-IFS model**, *JCGéo 2023*, Paris, France, <https://jcgeo.sciencesconf.org/>.
Boris Bordeaux, Iván Rasskin, Christian Gentil, Jorge Luis Ramírez Alfonsín, Lionel Garnier.
- March 2023 **Representation of polytopal sphere packings using the BC-IFS model**, *GTMG 2023*, Strasbourg, France, <https://gtmg2023.sciencesconf.org/>.
Boris Bordeaux, Iván Rasskin, Christian Gentil, Jorge Luis Ramírez Alfonsín, Lionel Garnier
- March 2022 **Construction of 2D lacunar transitions**, *GTMG 2022*, Dijon, France, <https://gtmg2022.sciencesconf.org/>.
Boris Bordeaux, Christian Gentil, Lionel Garnier

Seminars

- April 2026 **Automatic design of lacunar fractal structures**, *LIS*, Marseille, France, Seminar in the LIS lab on my PhD work and some recent results in the field.
Boris Bordeaux, Christian Gentil, Lionel Garnier, Charles-Adèle Desormeaux, Iván Rasskin
- May 2023 **Automatic design of fractal structures using the BC-IFS model**, *LIB*, Dijon, France, Seminar in the LIB lab on my PhD work.
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Posters

- March 2025 **Automatic design of lacunar fractal structures**, *PhD Students' Day*, Dijon, France, Presentation of PhD work to Bachelor's and Master's students, <https://lib.ube.fr/journee-des-doctorants-lib/>.
Boris Bordeaux, Christian Gentil, Iván Rasskin, Lionel Garnier

- March 2024 **Representation of polytopal sphere packings using the BC-IFS model**, *JNIM 2024*, Grenoble, France, National Days of Mathematical Computing, <https://jnim2024.sciencesconf.org/>.
Boris Bordeaux, Christian Gentil, Lionel Garnier, Iván Rasskin, Jorge Luis Ramírez Alfonsín
- February 2023 **Design of lacunar fractal structures**, *HCERES evaluation of LIB*, Dijon, France, Presentation of PhD research, <https://www.hceres.fr/fr/rechercher-une-publication/lib-laboratoire-dinformatique-de-bourgogne>.
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Award

- March 2023 **Best Presentation Award**, *GTMG 2023*, Strasbourg, France.
<https://gtmg2023.sciencesconf.org/>

Administrative Responsibilities

- 2023–24 **Alternate PhD Student Representative**, *LIB*, Dijon, France.
Organisation of events for PhD students in the lab
- 2023–25 **Co-organiser of the Young Researchers in Geometry Days**.
Organisation of annual events bringing together PhD students and postdocs in geometry:
2023: <https://jcgeo.sciencesconf.org/>
2024: <https://jcgeo24.sciencesconf.org/>
2025: <https://jcgeo25.sciencesconf.org/>

Science Outreach

- March 2024 **Research Meetings**, *UBE, Polytech Dijon*, Dijon, France.
- March 2025 Presentations to Polytech Dijon students about research topics and the PhD program.
- March 2024 **PhD Students' Day**, *UBE, Department of Computer Science, Electronics and Mechanics*, Dijon, France.
- March 2025 Presentations to students on research carried out by PhD students from the ImVia and LIB laboratories.
- March 2024 **Careers in Imaging**, *UBE, Department of Computer Science, Electronics and Mechanics*, Dijon, France.
Presentations to Bachelor's and Master's students about imaging research and the PhD path.
- June 2023 **Mathematics Rally Super-Final**, *UBE, Research Institute on Mathematics Education*, Dijon, France.
Workshop on fractals for middle school pupils.
<https://irem.ube.fr/le-rallye-mathematique-des-colleges-de-bourgogne/>
- October 2022 **Science Festival**, *UBE*, Dijon, France.
Public exhibition presenting LIB's research on fractals.
<https://www.fetedelascience.fr/conception-de-formes-bio-inspirees-pour-l-economie-d-energie>

Research Mobilities

- June 2023 **Young Researchers School in Mathematical Computer Science**, *XLIM*, University of Poitiers, France.
Five-day research school covering advanced topics in mathematical computer science, including combinatorial maps, graph transformations, mathematical morphology, and topological geometric modeling.
<https://ejcim23.sciencesconf.org/>
- December 2022 **Research stay**, *IMAG*, University of Montpellier, France.
One-week research stay on the topology of sphere packings, funded by the GdR IG-RV.
<https://gdr-igrv.fr/post/23-04-28-retours-mobilites/>

2024 – 2025 (Teaching and Research Fellow – ATER, 275h)

Level	Module	Type	Hours
1st-year undergraduate	Web Development (HTML, CSS, JS)	Tutorials	8h
		Labs	49h
1st-year undergraduate	Java Programming (Introduction and Algorithms)	Labs	60h
1st-year undergraduate	Object-Oriented Programming (Java)	Labs	72h
2nd-year undergraduate	Computer Graphics (ThreeJS)	Tutorials	2h
		Labs	20h
3rd-year undergraduate	Software Engineering (Git, Project Management, Object-Oriented Programming)	Tutorials	16h
		Labs	48h

2023 – 2024 (Additional Teaching Activities, 64h)

Level	Module	Type	Hours
1st-year undergraduate	Web Development (HTML, CSS, JS)	Labs	36h
1st-year undergraduate	Object-Oriented Programming (Java)	Labs	20h
1st-year graduate	Algorithms and Complexity (OCaml)	Labs	6h
2nd-year graduate	Internship supervision: automatic generation of fractal bricks		2h

2022 – 2023 (Additional Teaching Activities, 64h)

Level	Module	Type	Hours
1st-year undergraduate	Web Development (HTML, CSS, JS)	Labs	36h
3rd-year undergraduate	Geometric Modelling (Blender)	Tutorials	10h
		Labs	6h
2nd-year graduate	Supervised project: generalised barycentric coordinates		2h

2021 – 2022 (Part-time Lecturer, 91h)

Level	Module	Type	Hours
1st-year undergraduate	Web Development (HTML, CSS, JS)	Labs	56h
1st-year undergraduate	Object-Oriented Programming (Java)	Labs	11h
3rd-year undergraduate	Geometric Modelling (Blender)	Tutorials	10h
		Labs	14h

Teaching-related Activities

Module	Activities
Web Development	Exam preparation and grading, projects grading
Object-Oriented Programming	Project demonstrations grading
Images for the Web	Exams grading, project demonstrations grading
Software Engineering	Project demonstrations grading
Computer Graphics	Project demonstrations grading

Skills

Programming

C++, Java, Python, C

Web

HTML5, CSS3, Javascript, PHP, SQL, TypeScript, Angular

Computer Graphics

OpenGL, Blender, ThreeJS, Irit

Development Tools

Qt Creator, Netbeans, JetBrains IDEs, Android Studio

Development & Infrastructure

Git, Networking, Raspberry Pi

Image & Video Processing

OpenCV, FFmpeg

Digital Fabrication

FDM 3D Printers, Orca Slicer

Languages

French (native), English (B2, TOEIC 2020: 790 points)